Ahsan Habib

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SUMMARY

Backend developer and final-year CSE student (CGPA 3.76/4.00). Single-handedly built the production backend of atmabiswas.org in PHP and MySQL, with session-based authentication, CSRF/XSS protection and a GitHub Actions CI/CD pipeline. Also develops RESTful APIs with Node.js, Express and MongoDB, and multithreaded programs in C.

EXPERIENCE

Backend Developer | BRACU Cultural Club, BRAC University, Dhaka — Aug 2025 – Mar 2026

- Developed the club website's entire backend in PHP; awarded a certificate signed by its faculty advisors

Backend Developer Intern (Paid) | ATMABISWAS (NGO), Remote — Jun 2024 – Sep 2025

- Engineered the backend of atmabiswas.org as the sole backend developer, using PHP and MySQL (PDO), including an admin dashboard for publishing blogs, notices and job postings with auto-generated job codes
- Secured the admin panel with session-based authentication, CSRF tokens and protected endpoints; validated uploads to block executable files and sanitized blog HTML with HTML Purifier to prevent XSS
- Automated deployments with a GitHub Actions CI/CD pipeline that pushes to Hostinger over FTPS

EDUCATION

B.Sc. in Computer Science and Engineering | BRAC University | CGPA **3.76/4.00** — Expected Fall 2026

- CGPA **3.95** in 300–400 level CSE courses; Vice Chancellor's List: Summer 2024, Fall 2025, Spring 2026
- Coursework: Data Structures, Algorithms, Databases, Operating Systems, Computer Networks, Software Engineering

Higher Secondary Certificate (HSC) | Notre Dame College, Dhaka | GPA **5.00/5.00** — 2021

TECHNICAL SKILLS

- **Languages:** Java, PHP, JavaScript, SQL, Python, C, C++, Kotlin, x86 Assembly
- **Backend & Databases:** Node.js, Express, RESTful APIs, MySQL (PDO), MongoDB (Mongoose), caching
- **Security & DevOps:** Authentication, CSRF/XSS prevention, input validation, Git, GitHub Actions (CI/CD)
- **Frontend & Mobile:** React, HTML, CSS, Tailwind CSS, Vite, Android (Kotlin, Jetpack Compose)
- **Systems & ML:** Multithreading (POSIX threads), Flex/Bison, Arduino, ESP32, PyTorch, LoRA, Whisper, Qwen2.5-VL

PROJECTS

BRACU Routine Builder | GitHub — Summer 2026

- Created a full-stack MERN web app for planning BRAC University class routines, with real-time conflict detection that blocks overlapping classes, paired-day scheduling and two-slot lab spanning
- Designed a **19-endpoint** RESTful API with Express and Mongoose, including cascading deletes and auto-save
- Integrated the university's USIS course feed with a **7-day** cache to auto-fill sections, faculty, rooms and exam dates

Technologies: React, Node.js, Express, MongoDB, Mongoose

Process Manager Simulator | Team of 3 | GitHub — Spring 2026

- Owned the process-lifecycle module (fork, exit, wait, kill) and orphan reparenting on a mutex-protected process table, including a blocking wait free of missed-wakeup races

Technologies: C, POSIX threads

C-Subset Compiler | GitHub — Summer 2026

- Built a compiler front end for a C subset: Flex/Bison lexer and parser, scope-aware symbol table with hashed scopes, semantic analysis for type, scope and call errors, and a two-pass AST walk that emits three-address code

Technologies: C++, Flex, Bison

TrackAlert Vision: Patrol Robot | GitHub — Summer 2026

- **Led a team of 5** as lead architect: designed a two-controller system (Arduino Mega for real-time control, ESP32-S3-CAM for vision and Wi-Fi) and implemented the PD line follower and a Telegram photo-alert bot

Technologies: C++, Arduino Mega, ESP32-S3-CAM

Other projects: Bigger Number Game (Android) · Voting Machine (8086 Assembly) · Arduino Tic-Tac-Toe (C++)

RESEARCH

InsightLens: Lecture-to-Notes Pipeline for Code-Mixed Lectures | Undergraduate thesis, team of 3 — 2026

- Fine-tuned Whisper with LoRA on Bengali-English lectures, cutting character error rate from **67.7% to 15.8%**
- Developed the whiteboard-reading stage with Qwen2.5-VL, raising board-content recall from **31.2% to 88.8%**
- Graded **A (95.5/100)**; co-authoring a manuscript on this work for an IEEE journal

Technologies: Python, PyTorch, LoRA, Whisper, Qwen2.5-VL